THE FLAT LOCUS OVER A COMPLETE DVR IS REPRESENTABLE BY A CLOSED IMMERSION OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an affine torsion-quotient construction, an etale-equivalence-relation gluing with monomorphism identifications, a descent of the closed immersion property, and a global universal property argument by etale covering, all inside the delivered span

Editorial note. The label lemma-flat-locus names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Flatness over a complete DVR detected by vanishing of uniformizer-power torsion
- The presentation definition of locally Noetherian formal algebraic spaces, with etale transition maps between affines
- Descent of representability and of the closed immersion property along etale surjective morphisms, from the spaces and bootstrap lemmas cited

Source reference. Original lemma 046J: sources/stacks-chow/tags.txt, source lines 5474–5474. Prerequisite pointer invoked at proof line(s) 5816; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at proof line(s) 5774; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AIG: sources/stacks-chow/tags.txt, source lines 13264–13264. Prerequisite pointer invoked at proof line(s) 5790; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCG: sources/stacks-restricted/pinned-original-source.tex, source lines 3136–3136. Prerequisite pointer invoked at proof line(s) 5793; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 03KD: sources/stacks-chow/tags.txt, source lines 4673–4673. Prerequisite pointer invoked at proof line(s) 5818; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 02WW: sources/stacks-chow/tags.txt, source lines 3819–3819. Prerequisite pointer invoked at proof line(s) 5805; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a locally Noetherian formal algebraic space over a complete discrete valuation ring A. Then there exists a closed immersion $X' \to X$ of formal algebraic spaces such that X' is flat over A and such that any morphism $Y \to X$ of locally Noetherian formal algebraic spaces with Y flat over A factors through X'.*

Proof. Let $\pi \in A$ be the uniformizer. Recall that an A-module is flat if and only if the π-power torsion is 0.

First assume that X is an affine formal algebraic space. Then $X = \mathrm{Spf}(B)$ with B an adic Noetherian A-algebra. In this case we set $X' = \mathrm{Spf}(B')$ where $B' = B/\pi$-power torsion. It is clear that X' is flat over A and that $X' \to X$ is a closed immersion. Let $g : Y \to X$ be a morphism of locally Noetherian formal algebraic spaces with Y flat over A. Choose a covering $\{Y_j \to Y\}$ as in Formal Spaces, Definition 0AIM. Then $Y_j = \mathrm{Spf}(C_j)$ with C_j flat over A. Hence the morphism $Y_j \to X$, which correspond to a continuous R-algebra map $B \to C_j$, factors through X' as clearly $B \to C_j$ kills the π-power torsion. Since $\{Y_j \to Y\}$ is a covering and since $X' \to X$ is a monomorphism, we conclude that g factors through X'.

Let X and $\{X_i \to X\}_{i \in I}$ be as in Formal Spaces, Definition 0AIM. For each i let $X'_i \to X_i$ be the flat part as constructed above. For $i, j \in I$ the projection $X'_i \times_X X_j \to X'_i$ is an étale (by assumption) morphism of schemes (by Formal Spaces, Lemma 0AIG). Hence $X'_i \times_X X_j$ is flat over A as morphisms representable by algebraic spaces and étale are flat (Lemma 0GCG). Thus the projection $X'_i \times_X X_j \to X_j$ factors through X'_j by the universal property. We conclude that

$$R_{ij} = X'_i \times_X X_j = X'_i \times_X X'_j = X_i \times_X X'_j$$

because the morphisms $X'_i \to X_i$ are injections of sheaves. Set $U = \coprod X'_i$, set $R = \coprod R_{ij}$, and denote $s, t : R \to U$ the two projections. As a sheaf $R = U \times_X U$ and s and t are étale. Then $(t, s) : R \to U$ defines an étale equivalence relation by our observations above. Thus $X' = U/R$ is an algebraic space by Spaces, Theorem 02WW. By construction the diagram

$$\begin{array}{ccc} \coprod X'_i & \longrightarrow & \coprod X_i \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

is cartesian. Since the right vertical arrow is étale surjective and the top horizontal arrow is representable and a closed immersion we conclude that $X' \to X$ is representable by Bootstrap, Lemma 046J. Then we can use Spaces, Lemma 03KD to conclude that $X' \to X$ is a closed immersion.

Finally, suppose that $Y \to X$ is a morphism with Y a locally Noetherian formal algebraic space flat over A. Then each $X_i \times_X Y$ is étale over Y and therefore flat over A (see above). Then $X_i \times_X Y \to X_i$ factors through X'_i. Hence $Y \to X$ factors through X' because $\{X_i \times_X Y \to Y\}$ is an étale covering. □